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Short communication

Unique behavior of dermal cells from regenerative mammal, the African Spiny Mouse, in response to substrate stiffness

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ABSTRACT

The African Spiny Mouse (*Acomys* spp.) is a unique outbred mammal capable of full, scar-free skin regeneration. *In vivo*, we have observed rapid reepithelialization and deposition of normal dermis in *Acomys* after wounding. *Acomys* skin also has a lower modulus and lower elastic energy storage than normal lab mice, *Mus musculus*. To see if the different *in vivo* mechanical microenvironments retained an effect on dermal cells and contributed to regenerative behavior, we examined isolated keratinocytes in response to physical wounding and fibroblasts in response to varying substrate stiffness. Classic mechanobiology paradigms suggest stiffer substrates will promote myofibroblast activation, but we do not see this in *Acomys* dermal fibroblasts (DFs). Though *Mus* DFs increase organization of α -smooth muscle actin (α SMA)-positive stress fibers as substrate stiffness increases, *Acomys* DFs assemble very few α SMA-positive stress fibers upon changes in substrate stiffness. *Acomys* DFs generate lower traction forces than *Mus* DFs on pliable surfaces, and *Acomys* DFs produce and modify matrix proteins differently than *Mus* in 2D and 3D culture systems. In contrast to *Acomys* DFs “relaxed” behavior, we found that freshly isolated *Acomys* keratinocytes retain the ability to close wounds faster than *Mus* in an *in vitro* scratch assay. Taken together, these preliminary observations suggest that *Acomys* dermal cells retain unique biophysical properties *in vitro* that may reflect their altered *in vivo* mechanical microenvironment and may promote scar-free wound healing.

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1. Introduction

The African Spiny Mouse (*Acomys* spp.) is a mammal with remarkable regenerative abilities. Following full-thickness skin removal, *Acomys* regenerates in a scar-free manner and replaces dermis, hairs, smooth muscle of the erector pili muscles, sebaceous glands, skeletal muscle of the panniculus carnosus, and adipose cells (Brant et al., 2015, 2016; Seifert et al., 2012). *Acomys* can also regenerate cartilage after an ear punch (Gawriluk et al., 2016; Seifert et al., 2012) and restore cardiac function after myocardial infarction (Qi et al., 2017, 2016). To translate this great potential for regeneration to other mammals, the cellular and molecular basis of *Acomys* scar-free healing must be established.

We previously characterized events of skin regeneration *in vivo* and identified several differences between regenerating *Acomys* and scarring *Mus* skin. Notably, full-thickness *Acomys* skin has a 20× lower modulus and 70× lower toughness than *Mus* skin (Seifert et al., 2012). Since hard surfaces are known to promote myofibroblast activity (Hinz, 2010), we hypothesized that *Acomys* cells isolated from softer microenvironments may maintain an inactivated fibroblast phenotype more readily than *Mus* cells isolated from rigid microenvironments. We were also interested in keratinocytes since stiffer environments often accelerate migration but, conversely, rapid reepithelialization of *in vivo* wounds has been observed in *Acomys*. Here, we have analyzed epidermal keratinocytes and dermal fibroblasts from *Acomys* and *Mus* to determine functional differences between the two cell types that may contribute to dermal wound healing.

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2. Materials and methods

2.1. Isolation and culture of primary cells

Protocols and care of *Acomys cahirinus* (University of Florida colony) and outbred CD-1 *Mus musculus* (Charles River) were approved by UF Institutional Animal Care and Use Committee and were within US animal welfare regulations and guidelines.

Cells that had gone through less than 3 population doublings after isolation from newborn pups of *Acomys* (5-week gestation) and *Mus* (3-week gestation) were used for these experiments. Birth is well beyond the stage when embryos can heal without scarring *in utero* (embryonic day 16.5 in *Mus*), thus fibroblasts were obtained from “scarring” stages. Pups were euthanized and the dorsal skin removed. Dermis and epidermis were separated after overnight incubation at 4 °C in 0.125% Trypsin with EDTA (Gibco). The dermis was further incubated in 0.1% collagenase type-1 (Gibco) for 1.5 h at 37 °C to separate cells and then washed and cultured in DMEM with 10% FBS, 10% NuSerum (Corning), 0.1% insulin-transferrin-selenium, and 0.1% penicillin/streptomycin. The epidermis was triturated and filtered, and isolated cells were cultured in keratinocyte-specific medium (Lifeline® DermaK) supplemented with pen/strep.

2.2. In vitro wound healing assay

Primary keratinocytes were seeded at 2×10^5 cells per well in 24-well plates overnight. Cell layer was scratched using a sterile pipette tip and imaged directly after scratching. Images of the same scratched region were taken with phase contrast every three hours. The rate of closure was calculated using ImageJ (NIH).

2.3. Silicone substrate fabrication

Polydimethylsiloxane (PDMS, Sylgard527 and Sylgard184, Dow Corning) was mixed per manufacturer's instructions, poured into a tissue culture plate, degassed, and cured at 50 °C. Surfaces were plasma treated (PDC-001-HP, Harrick) for 25 s and submerged in deionized water (DIW). Samples were rinsed, covered with sterile DIW, and sterilized by germicidal UV. PDMS and plastic dishes were coated with 10 µg/mL rat-tail collagen I (Coll, Corning) solution.

2.4. Immunofluorescence and Western Blot

Acomys and *Mus* dermal fibroblasts (DFs) were plated onto PDMS and plastic surfaces at 5000 cells/cm² in normal serum-containing medium for 48 h. To preserve the F-actin/ α SMA structure, cells were treated with 3% paraformaldehyde (PFA) and 0.01% Triton-X100 for 5 min, further fixed for 10 min in 3% PFA, and blocked in 1% BSA for 30 min. Cells were then labeled for F-actin (Phalloidin-FITC, 1:150 in 1% BSA, Sigma) for 40 min and α SMA (α SMA-Cy3 monoclonal antibody, 1:200 in 1% BSA, Sigma#C6198) for 1 h. Analyzing our preliminary *Acomys* genome assembly, α SMA shows 99.4% identity with *Mus* α SMA. Additionally, the antibody used cross-reacts with 12 animals including canines, humans, frogs, and mice. Images were acquired (Nikon Ni-Eclipse) with a 40× immersion objective. All images were taken at identical settings for comparison and subsequent quantification in MATLAB R2016a (Mathworks). Statistics were performed in JMP Pro 13 (SAS).

For Western Blot, cell extracts from DFs on plastic were made with RIPA buffer, and equal amounts of protein were loaded onto a 4–12% bis-tris gel and blotted onto a PVDF membrane. α SMA antibody (Abcam#5694) with high-species cross-reactivity was

used. The membrane was incubated with 1:10,000 dilution of α SMA antibody or GAPDH made up in 5% BSA in TBST overnight, then with an HRP secondary antibody at 1:1000. Bands were visualized with a chemiluminescence kit and imaged on a FluorChem imager.

To visualize ECM production, primary DFs were seeded at 15×10^4 in untreated 24-well plates, grown to confluency, washed with PBS, fixed for 30 min in 4% PFA, permeabilized with 0.1% TX-100, and blocked with 0.5% BSA. Individual wells were incubated with Coll, CollIII, or CollIV primary antibodies (1:100; Abcam) followed with AlexaFluor488-conjugated anti-rabbit or anti-mouse IgG (1:500; Abcam) and imaged using Olympus IX81 40X objective.

2.5. Traction Force Microscopy (TFM)

Two-layered polyacrylamide (PA) hydrogel substrates ($E \sim 50$ kPa) were fabricated for TFM as in Simmons et al. (2013). Gel surface was SulfoSANPAH functionalized, sterilized by germicidal UV, and coated with 100 µg/mL Coll for 30 min before seeding 10,000 cells per substrate. Cells were allowed to attach for 18–24 h before transfer to a temperature-controlled stage (Nikon Ti-E). One brightfield image of the attached cell was acquired with a fluorescence image of beads. Cell locations were stored (Nikon Elements) then cells removed using trypsin, and second image of “null” beads acquired. Calculations of bead displacements (PIVLab (Thielicke and Stamhuis, 2014)) and strain energy (Fourier-transform traction cytometry (Sabass et al., 2008)) were done in MATLAB. Area and roundness were quantified by manually outlining cell border and running “Measure” in ImageJ, where roundness = $\frac{4 \cdot \text{Area}}{\pi \cdot \text{Major Axis}^2}$. Wilcoxon non-parametric tests were used to compare calculated strain energy (one-sided test) and cell area and roundness (two-sided test) in JMP Pro 13.

To demonstrate fidelity of mechanism for cell-generated traction forces, *Mus* and *Acomys* DFs were plated on Coll-coated compliant PDMS substrates ($E \sim 5$ kPa compared to $E \sim 50$ kPa for PA hydrogels). After 24 h in serum-containing medium, two brightfield images (2.31 mm²) from two wells of each species were taken. Wrinkling was assessed by thresholding images and counting pixels in ImageJ.

2.6. Cell-embedded collagen hydrogels

High concentration Coll was diluted with 0.2% acetic acid and combined in a 3:1 ratio with 5× DMEM (Sigma), 1 M HEPES (Gibco), and DFs to fabricate 3 mg/mL collagen hydrogels with 2000 cells per 55 µL gel. Precursor solution is maintained at 4 °C and then incubated at 37 °C for 30 min. Constructs are then hydrated with media and kept in 37 °C CO₂ incubator. Gels were fixed with 4% PFA in PBS and later lyophilized using a critical point drier (Tousimis, autosamdri-815). Samples were coated with carbon using a sputter coater (Denton DeskV) and imaged with a Scanning Electron Microscope (Hitachi SU5000).

3. Results and discussion

3.1. *Acomys cahirinus* keratinocytes close in vitro wounds faster than *Mus musculus*

In vivo, *Acomys* appear to reepithelialize wounds more quickly than *Mus*, but scabs can obscure visualization of epithelial migration and calculation of wound closure rates. *In vitro*, wound healing rates were more than twice as fast for *Acomys* keratinocytes (AKs, 0.021 mm²/h) than *Mus* keratinocytes (MK, 0.009 mm²/h). AKs closed scratches under 30 h whereas most MK scratches were still

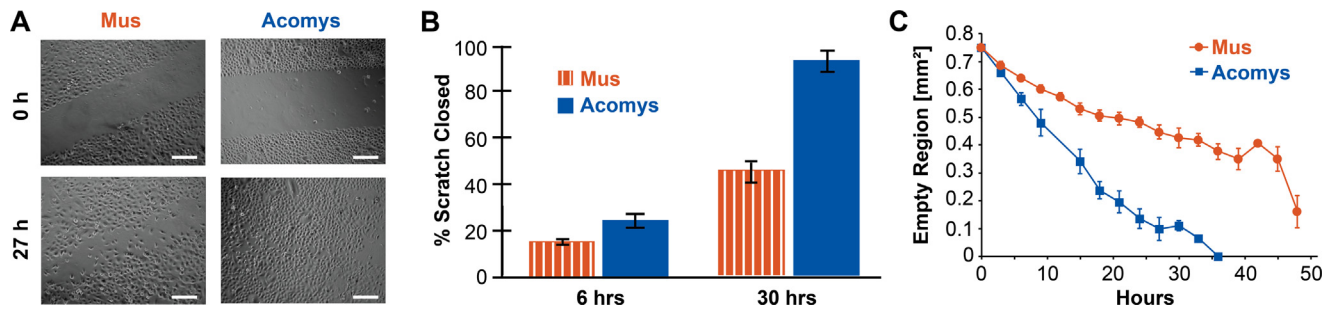


Fig. 1. *Acomys cahirinus* keratinocytes (AKs) migrate faster than *Mus musculus* keratinocytes (MKs) during wound closure. (A) Sample images of keratinocyte cultures at 0 h and 27 h after induced scratch. Scale bars are 200 μ m. (B–C) AKs migrated and closed the scratch area within 30 h while MKs require more than 48 h to close completely. Bars depict mean $\pm$ standard error, $n \geq 8$ for each condition.

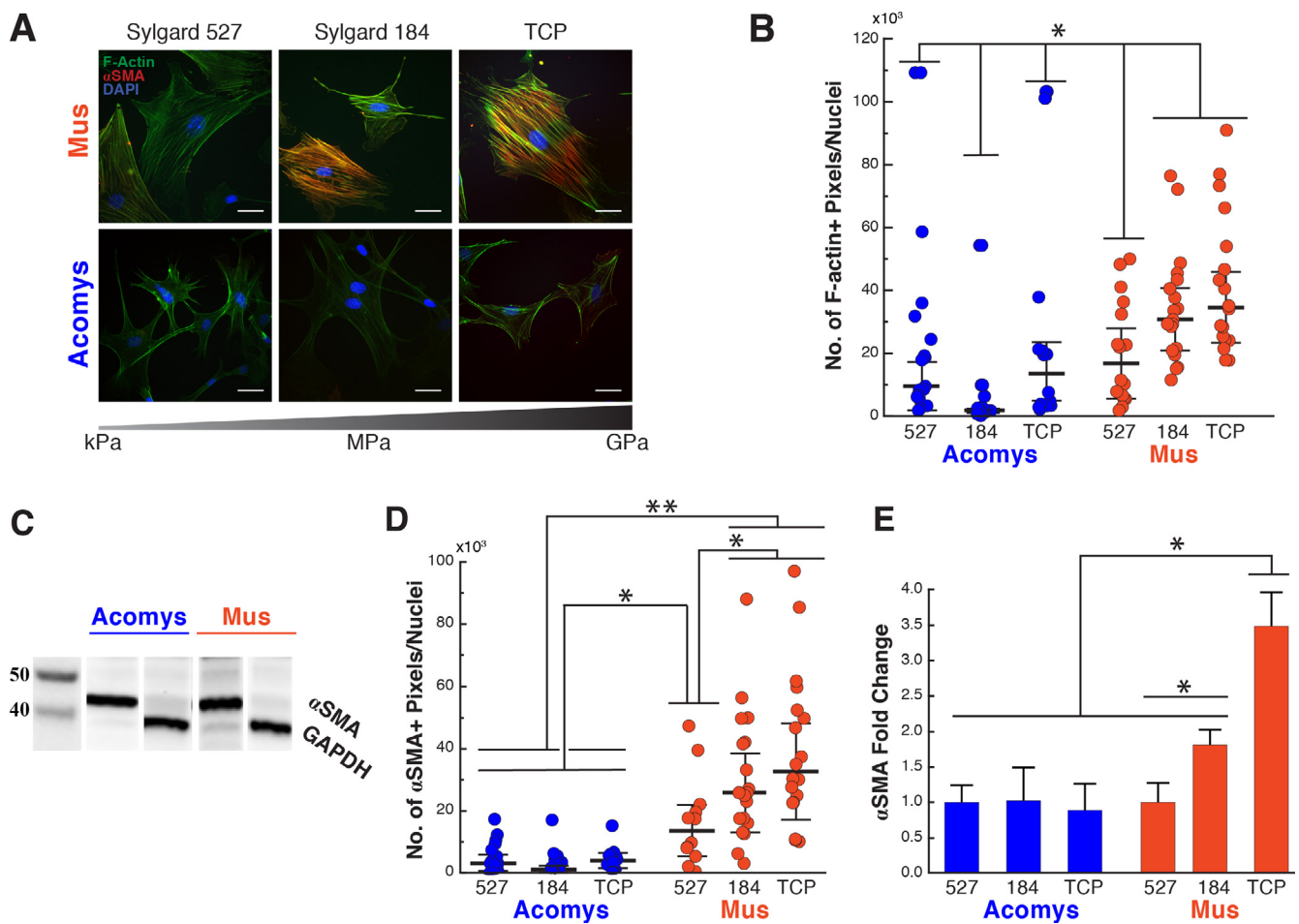


Fig. 2. Substrate stiffness does not upregulate α SMA expression in *Acomys* dermal fibroblasts (ADFs). (A) *Mus* dermal fibroblasts (MDFs) upregulate α SMA (red) proportionally on silicone substrates with increasing stiffness (top row), while ADFs did not show any significant changes in α SMA expression (bottom row). Both cell types assemble F-actin fibers (green), though MDFs assemble more vivid, discrete fibers on higher stiffness substrates. Scale bars are 25 μ m. (B) Quantification of F-actin-positive pixels normalized to number of nuclei confirms ADFs spread and form an F-actin cytoskeleton, but no significant increase is seen in ADFs in response to stiffness compared to MDFs ($n = 46$ for *Acomys*, $n = 53$ for *Mus*). Bars depict median $\pm$ median absolute deviation. (C) Western blot confirms α SMA (42 kDa) is produced by ADFs on plastic and recognized by commercial antibodies. Equal amounts of protein including GAPDH 37 kDa control were run in each lane for both species on a single gel. Molecular weight standards (kDa) shown in left column. (D) Quantified pixel count of α SMA-positive pixels normalized to number of nuclei shows ADFs (blue data points) do not assemble as much α SMA as MDFs (orange data points) in response to stiffness. No significant difference is seen in α SMA production per nuclei between ADFs on different stiffnesses, while MDFs had significantly more α SMA with increasing stiffness compared to ADFs. Bars depict median $\pm$ median absolute deviation. (E) ADFs do not have significant fold-change between stiffnesses while MDFs have 1.5 $\times$ increase (Sylgard 184, $E \sim 1$ MPa) and 3 $\times$ increase (TCP, $E \sim 3$ GPa) in α SMA as substrate stiffness increases. Fold-change was determined by normalizing to the average number of α SMA-positive pixels on Sylgard 527 ($E \sim 5$ kPa). No significant difference was seen between ADFs on all stiffnesses or MDFs on Sylgard 527. Bars depict mean $\pm$ standard error. Statistics were determined using a Wilcoxon non-parametric multiple comparison test. * $p < 0.03$, ** $p < 0.0001$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

detectable 48 h later (Fig. 1). This *in vitro* migration behavior is consistent with *in vivo* observations (Seifert et al., 2012).

3.2. α SMA expression in *Acomys* dermal fibroblasts does not change with substrate stiffness

During wound healing, fibroblasts become activated myofibroblasts that produce inflammatory factors and assemble dense extracellular matrix (ECM). One sign of myofibroblast activation is increased α SMA expression as fibroblasts generate higher contractile forces to close wounds (Hinz, 2010), and such α SMA expression has been induced *in vitro* by culturing fibroblasts on substrates with hyper-physiological stiffness (Achterberg et al., 2014; Goffin et al., 2006; Quinlan and Billiar, 2012; Scott et al., 2017; Wang et al., 2013). Since *Acomys* does not have activated myofibroblasts *in vivo* after wounding and *Acomys* skin is softer than *Mus* (Seifert et al., 2012), we hypothesized that *Acomys* DFs would not assemble α SMA-positive stress fibers in response to stiff substrates.

We did observe less α SMA assembly in *Acomys* fibroblasts (ADFs) than *Mus* fibroblasts (MDFs) on substrates of increasing stiffness. MDFs had clear, organized α SMA-positive fibers that increased proportionally with substrate stiffness (Fig. 2A), similar to previous findings (Achterberg et al., 2014; Goffin et al., 2006). In contrast, we saw no significant difference in α SMA localization in ADFs between substrates (Fig. 2A, D–E). Quantification of α SMA and F-actin fluorescence showed that ADFs had lower α SMA values compared to MDFs that remained constant across stiffnesses, and we confirmed by Western blot that ADFs on plastic produce α SMA (Fig. 2C). Both DFs assembled F-actin fibers, though F-actin fibers were more pronounced in MDFs (Fig. 2B). These data suggest that

ADFs cannot be activated into myofibroblasts by stiff substrates alone. Since *Acomys* gestation takes longer than *Mus*, *Acomys* newborn fibroblasts are actually “older” than *Mus*, which accentuates this surprising response of ADFs.

3.3. *Acomys* fibroblasts generate lower contractile forces than *Mus* fibroblasts

Since α SMA enhances but is not required for contractile force generation (Chen et al., 2007), we sought to quantify ADF and MDF traction forces and hypothesized that ADFs would generate lower cellular traction forces than MDFs. ADFs did generate lower traction forces than MDFs on PA substrates (1.0 ± 0.6 pJ versus 2.4 ± 1.8 pJ, $p = 0.04$, $n = 12$ each). Median $\pm$ median absolute deviation reported to reflect non-parametric statistical analysis. Cell spread area and roundness were similar between cell populations (Fig. 3B–C), so calculated differences likely reflect true differences in contractility and not cell shape. To examine differences in traction force on different substrates, wrinkle analysis on ~ 5 kPa silicone confirmed greater contractility in *Mus* (Fig. 3F–H) compared to *Acomys*.

In normal mammalian fibroblasts, contractility is required for many processes central to fibrosis, including migration (Case and Waterman, 2015; Shi-wen et al., 2009), matrix protein production (Mun et al., 2014), and activation of TGF β , a pro-fibrotic cytokine (Mun et al., 2014; Wipff et al., 2007). Matrix stiffness has been shown to promote these fibrotic processes *in vitro*, suggesting lack of response to stiff substrates in *Acomys* may contribute to *Acomys* regeneration though exact mechanisms remain to be identified.

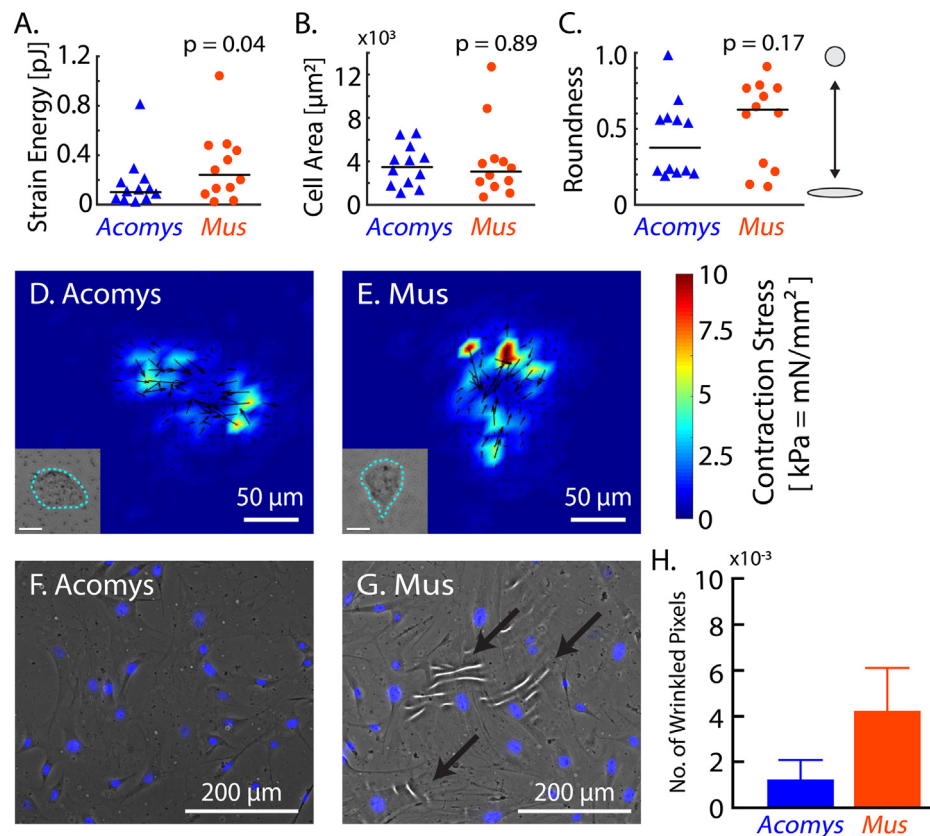


Fig. 3. *Acomys* dermal fibroblasts (ADFs) generate less traction energy than *Mus* dermal fibroblasts (MDFs) while having similar cell morphologies. (A) Strain energy represents work done per cell, and ADFs generated significantly less strain energy on polyacrylamide hydrogel surfaces than MDFs ($p = 0.04$). (B–C) Cells from both species had similar cell area and roundness. Black bars depict the median. (D–E) Traction stresses of representative ADFs and MDFs. (F–G) On compliant silicone substrates (Sylgard 527, $E \sim 5$ kPa), MDFs generate enough force to wrinkle the surface while ADFs do not. (H) Quantification of wrinkling (bright white areas) confirms MDFs deform the surface more than ADFs ($n = 4$ independent regions (2.31 mm^2) per species). Bars depict mean $\pm$ standard error. All scale bars are $50 \mu\text{m}$.

3.4. *Acomys* fibroblasts produce less matrix in 2D and 3D environments

In addition to increased α SMA expression and contractility, activated myofibroblasts produce and crosslink excess collagen. We then asked whether there were differences in fibroblast production of ECM proteins. Both MDFs and ADFs expressed ColI but with a greater intensity in MDFs (Fig. 4A–B). Only MDFs expressed

ColIII (Fig. 4C–D) and both cell types expressed ColIV (Fig. 4E–F). These results are somewhat surprising since ColIII is thought to be a regenerative collagen (Volk et al., 2011).

We also observed fibroblast remodeling in 3D. MDFs encapsulated in collagen spread and increase matrix stiffness over 7 days in culture (Fig. 4G, I), as expected from normal mammalian fibroblasts (see example review (Brown, 2013)). ADFs, on the other hand, do not spread nor remodel surrounding gel (Fig. 4H, J). When characterized by indentation after 7 days in culture, the MDF-embedded matrix was stiffer than the ADF-embedded matrix (1500 and 800 Pa, respectively).

4. Conclusion

The data presented here confirm the unique behavior of *Acomys* keratinocytes and DFs is conserved *in vitro*. We have shown that *Acomys* keratinocytes migrate faster than *Mus* keratinocytes to close an imposed wound region. We have also demonstrated that *Acomys* DFs do not adapt a myofibroblast phenotype *in vitro* as they fail to assemble α SMA-positive stress fibers in response to increasing substrate stiffness, generate lower contractile forces than their *Mus* counterparts, and do not produce excess collagen in 2D nor 3D compared to *Mus* DFs. Collectively, the behavior shown here suggests alterations in ADF mechanosensing pathways and provides the groundwork for extensive future investigations into mechanisms of mammalian regeneration.

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Conflict of interest

Authors disclose no financial and personal relationships with other people or organizations that could inappropriately bias their work.

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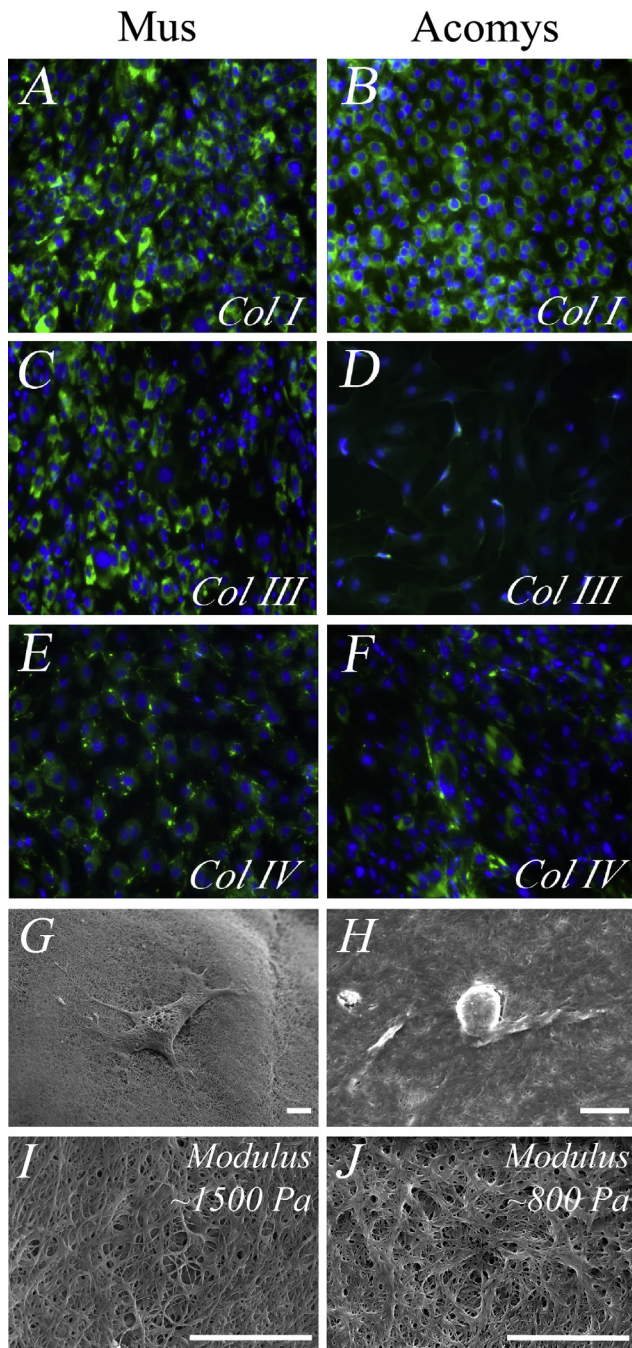


Fig. 4. *Acomys* dermal fibroblasts (ADFs) produce less matrix *in vitro* than *Mus* dermal fibroblasts (MDFs). ADFs and MDFs expressed collagen I (Col I, A, B) and collagen IV (Col IV, E, F) at similar levels, but MDFs expressed more collagen III (Col III) than ADFs (C, D). In 3D collagen I gels, MDFs are more stellate than spheroidal ADFs (G, H). Day 7 remodeled matrix of gels embedded with MDFs (I) are twice as stiff as those embedded with ADFs (J). Modulus values are nominal average of 3 indentations on 3 gels each (n = 9) using protocols from Rubiano et al. (2018). All scale bars are 10 μ m.

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